

Surface tension correlations from symbolic regression

$$T_r = \frac{T}{T_c}, \quad F_1 = 1 - T_r$$

Equation 1

Complexity: 1

Loss: 19.9917

$$\sigma = 7.611$$

Equation 2

Complexity: 3

Loss: 0.852208

$$\sigma = (1 - T_r)48.25$$

Equation 3

Complexity: 5

Loss: 0.00143133

$$\sigma = ((1 - T_r)26.19)^{1.351}$$

Equation 4

Complexity: 7

Loss: 0.000107575

$$\sigma = (((1 - T_r)29.08) - 0.1882)^{1.297}$$

Equation 5

Complexity: 9

Loss: 1.15259e-5

$$\sigma = (((1 - T_r)31.8) - 0.6746)^{1.26} + 0.4703$$

Equation 6

Complexity: 11

Loss: 1.13977e-5

$$\sigma = (((((1 - T_r) + (1 - T_r))15.94) - 0.6893)^{1.259} + 0.4835$$

Equation 7

Complexity: 15

Loss: 4.15658e-6

$$\sigma = \left((((1 - T_r) + (1 - T_r))((3.023)^{1-T_r} - 1.717) + 15.07)) - 0.06706 \right)^{1.375}$$

Equation 8

Complexity: 19

Loss: 3.02588e-6

$$\sigma = \left((((1 - T_r) + (1 - T_r))((1 - T_r)(1 - T_r)) + (((3.281)^{1-T_r} - 1.895) + 15.3))) - 0.07132 \right)^{1.376}$$

Equation 9

Complexity: 21

Loss: 1.64121e-7

$$\sigma = \left((((1 - T_r) + (1 - T_r))(((1 - T_r) - 2.058)(3.749)^{1 - T_r} + 15.9)) + ((1 - T_r) - 0.3) \right)^{1.333} + 0.1989$$